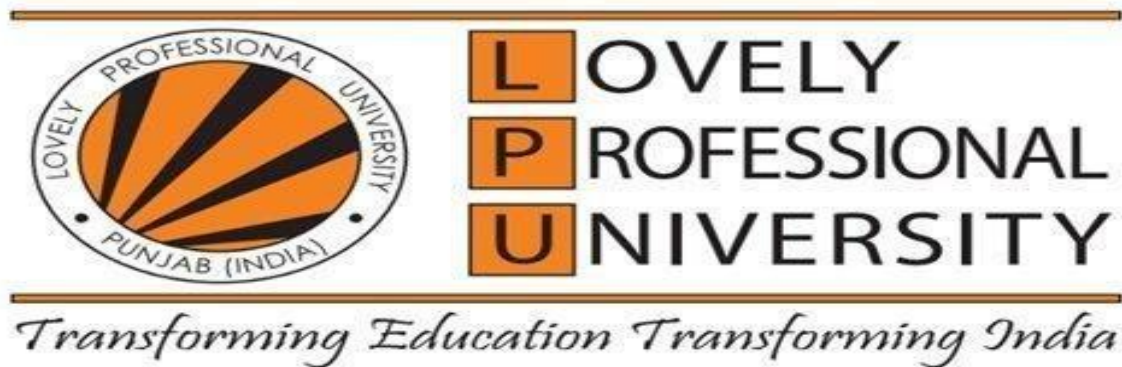




HOPEBRIDGE – CLOUD PROJECT REPORT

NGO Donation & Campaign Management Platform using Microsoft Azure

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1. INTRODUCTION

HopeBridge is a cloud-enabled NGO donation and campaign management platform developed to connect donors, NGOs, and administrators through a centralized system. The project uses Microsoft Azure cloud services for secure storage, identity management, network security, and monitoring.

Cloud computing enables organizations to access scalable, secure, and cost-effective computing resources over the internet. Modern applications use cloud platforms to improve storage management, monitoring, security, and deployment efficiency.

The HopeBridge project is a cloud-enabled NGO donation and campaign management platform developed using Node.js, Express.js, MongoDB, and Microsoft Azure services. The platform allows NGOs to create fundraising campaigns, upload campaign images, and receive donations while securely managing user access and cloud resources.

The project demonstrates the practical implementation of cloud computing concepts such as cloud storage, identity management, role-based access control, virtual networking, monitoring, and secure cloud resource management using Microsoft Azure.

2. BACKGROUND

Many NGOs face difficulties in managing donations, organizing campaigns, and securely storing campaign data. The HopeBridge platform was developed to simplify fundraising activities and improve transparency using cloud technologies.

Non-Governmental Organizations (NGOs) often face challenges in managing fundraising campaigns, donor interactions, and secure storage of campaign data. Traditional systems may suffer from scalability issues, poor accessibility, and lack of centralized monitoring.

Cloud computing solves these issues by providing scalable infrastructure, secure access management, and reliable storage services. Microsoft Azure offers multiple cloud solutions such as Azure Blob Storage, Azure Entra ID, Azure RBAC, Virtual Networks, and Monitoring services which help in building enterprise-grade applications.

The HopeBridge platform was developed to utilize these cloud technologies for creating a secure and efficient donation management system.

3. PROJECT DESCRIPTION

The platform allows NGOs to create fundraising campaigns, upload campaign images, manage users, and monitor activities. Azure Blob Storage is used for image storage, Azure Entra ID is used for identity management, and Azure Networking services are used for secure infrastructure.

HopeBridge is a web-based NGO donation and campaign management platform where NGOs can register, create campaigns, upload campaign images, and receive donations from donors.

Main Features:

- NGO Registration and Login
- Donor Access
- Campaign Creation and Management
- Image Upload using Azure Blob Storage
- Secure Role-Based Access Control (RBAC)
- Azure Entra ID Identity Management
- Virtual Network and Network Security Group Configuration
- Azure Monitoring and Metrics Tracking
- Admin Dashboard Management

Technologies Used:

- Frontend: EJS, HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB Atlas
- Cloud Platform: Microsoft Azure

4. CLOUD SERVICES USED

1. Azure Blob Storage

Used to store campaign images securely in the cloud.

2. Azure Entra ID

Used for identity and access management.

3. Azure Role-Based Access Control (RBAC)

Used to assign secure permissions to users.

4. Azure Virtual Network (VNet)

Used to create isolated cloud network infrastructure.

5. Network Security Group (NSG)

Used to control network traffic and improve security.

6. Azure Metrics Monitoring

Used to monitor storage transactions and cloud resource usage.

7. MongoDB Atlas

Used as the cloud database for storing application data.

SYSTEM ARCHITECTURE

User → Web Application → Node.js Server → MongoDB Atlas



Azure Blob Storage



Azure Monitoring & Security

The web application communicates with MongoDB Atlas for storing user and campaign data. Campaign images are uploaded to Azure Blob Storage. Azure Entra ID and RBAC provide secure identity and permission management. Azure Monitoring tracks cloud resource usage and transactions.

5. IMPLEMENTATION AND RESULTS

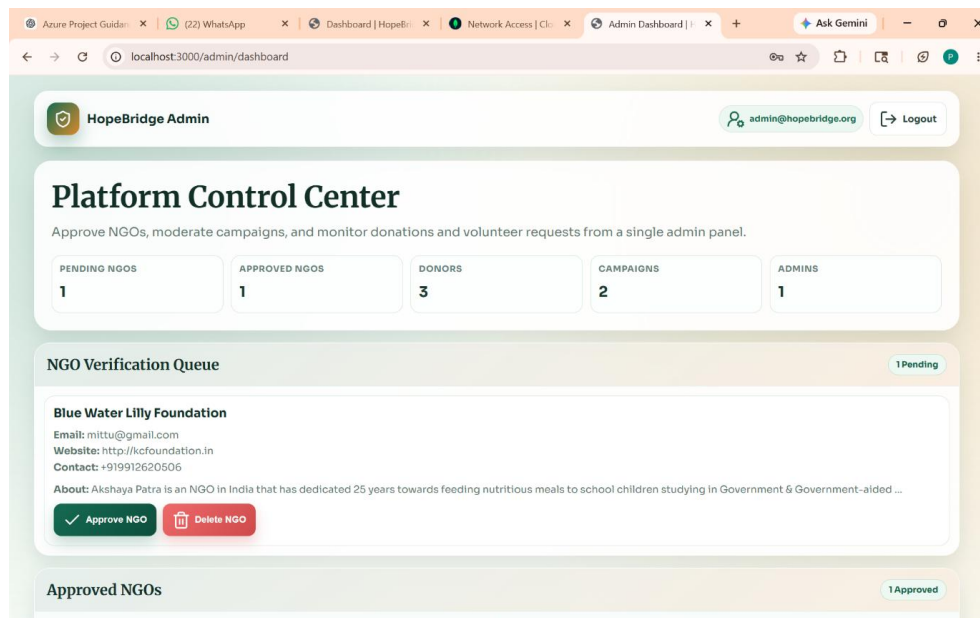


Figure 1: website Admin Dashboard

New Group - Microsoft Azure

Ask Gemini

portal.azure.com/?Microsoft_Azure_Education_correlationId=bf820e1e-1540-424a-8376-46378ae7a02d&Microsoft_Azure_Education_newA4...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

buddulapoorvireddy@ig...
DEFAULT DIRECTORY (BUDDULA...

Home > Default Directory | Overview

New Group

Got feedback?

Group type * ⓘ
Security

Group name * ⓘ
HopeBridge-Admins

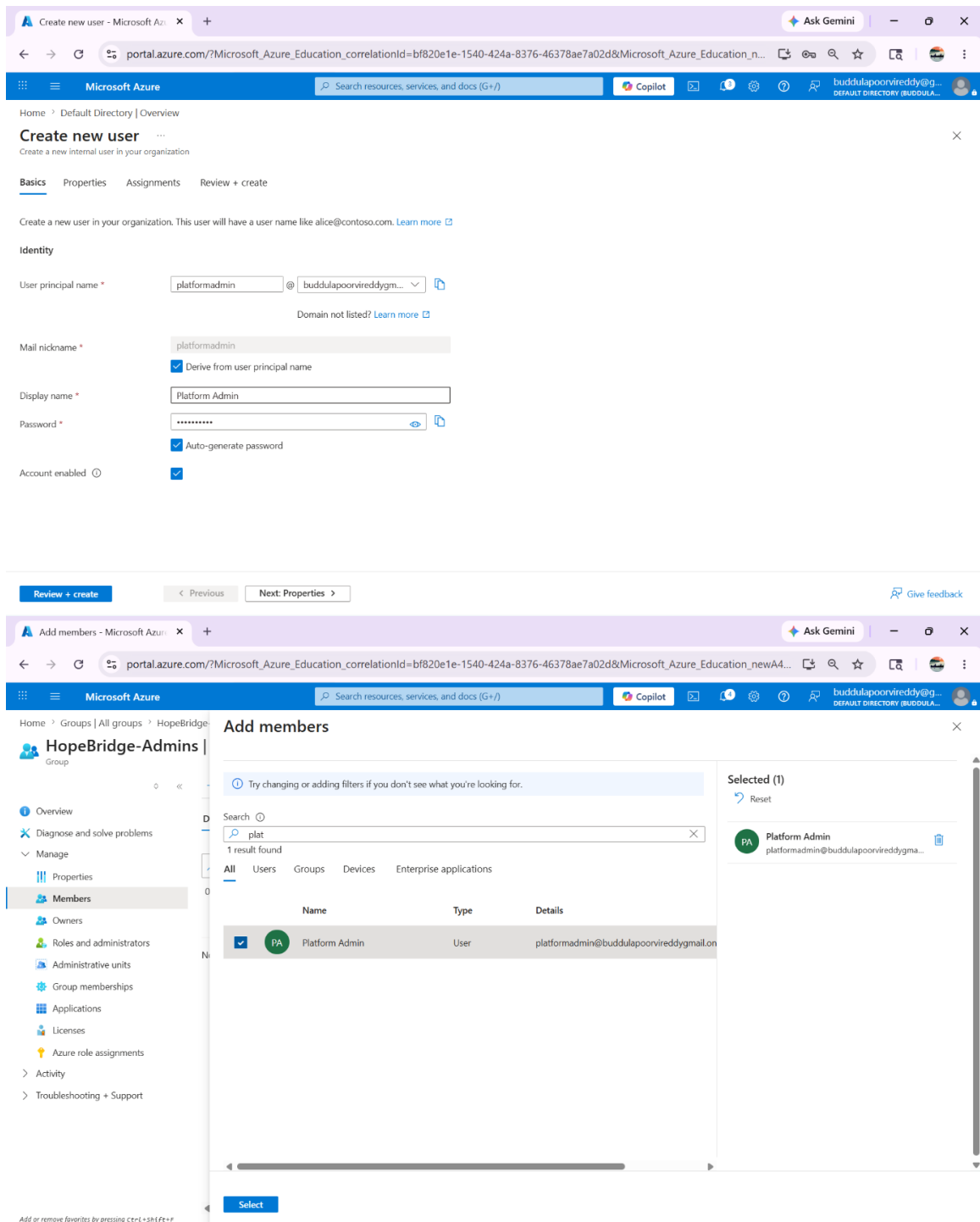
Group description ⓘ
Enter a description for the group

Membership type ⓘ
Assigned

Owners
No owners selected

Members
No members selected

Create



This figure demonstrates the implementation step related to creating azure storage account in the HopeBridge cloud project.

Here for ADMINS group read and RBAC role is assigned as OWNER

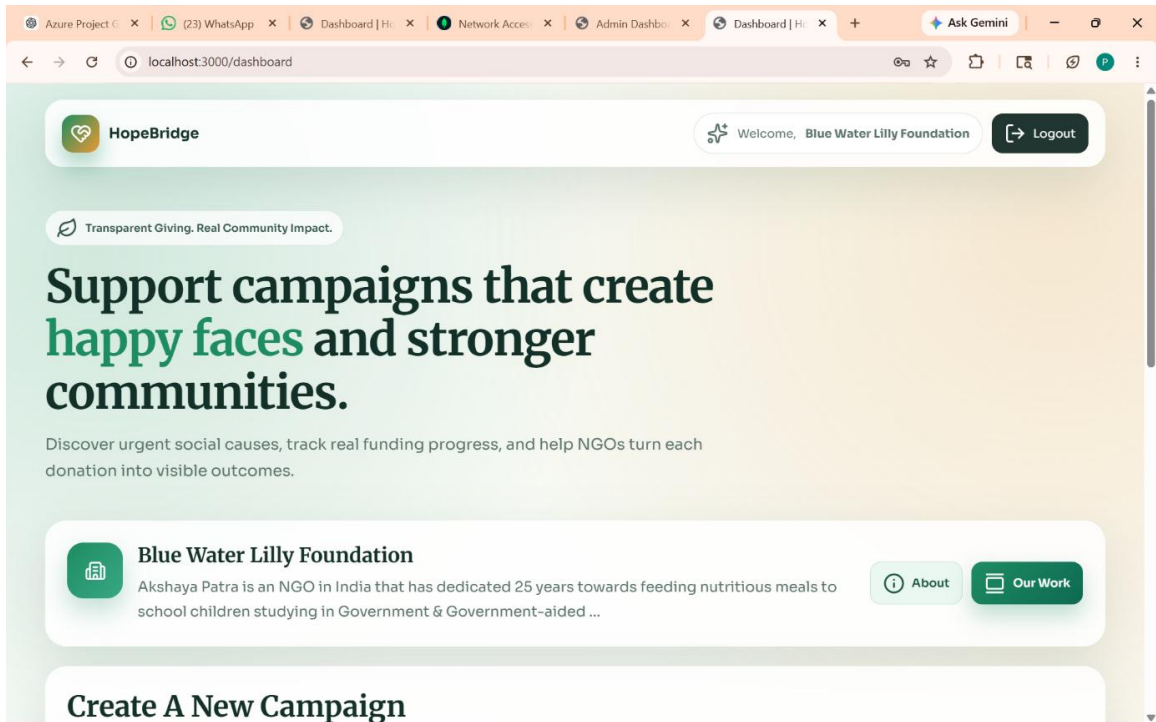
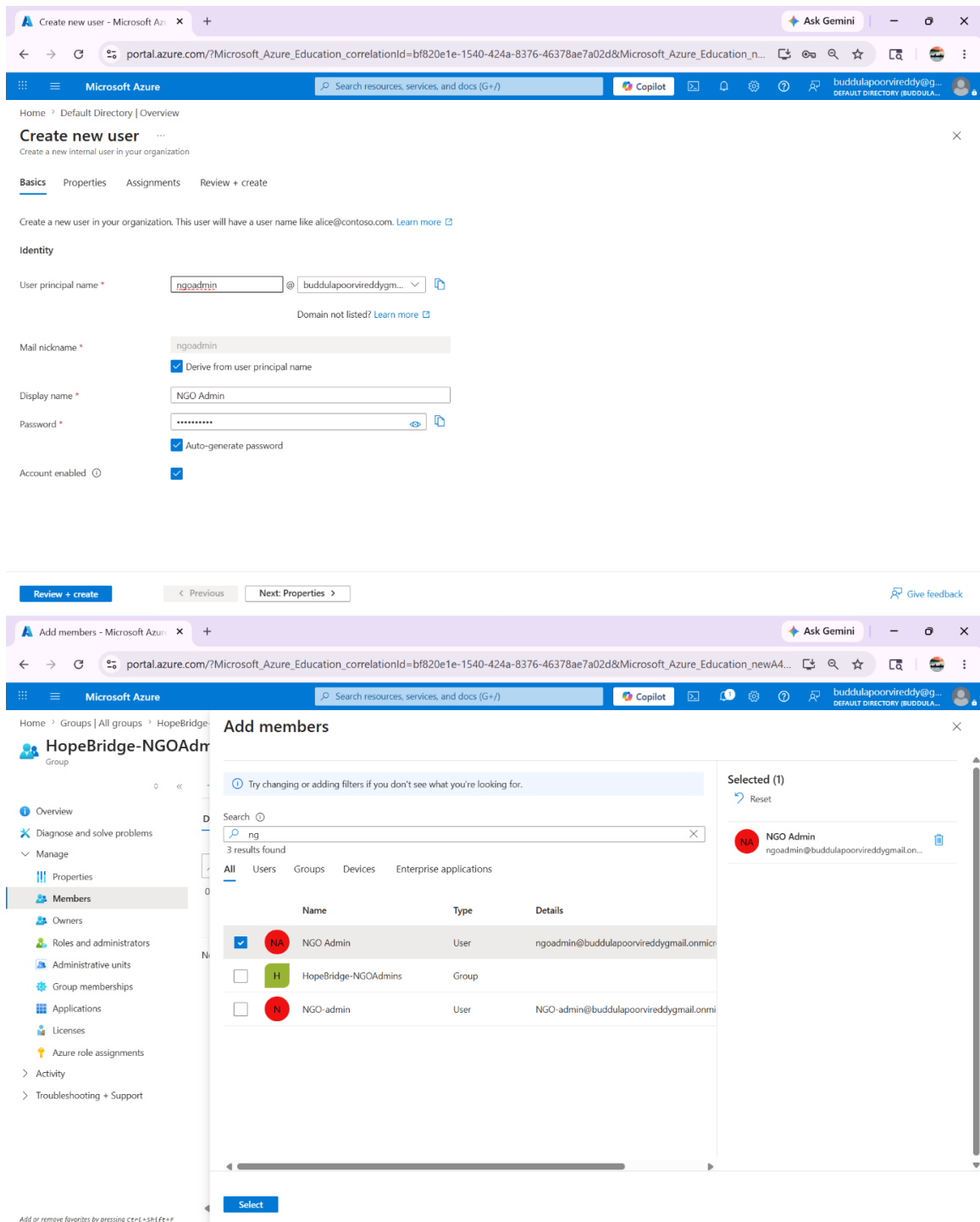


Figure 2:THIS dash board of NGO admin



This figure demonstrates the implementation step related to storage account deployment in the HopeBridge cloud project.

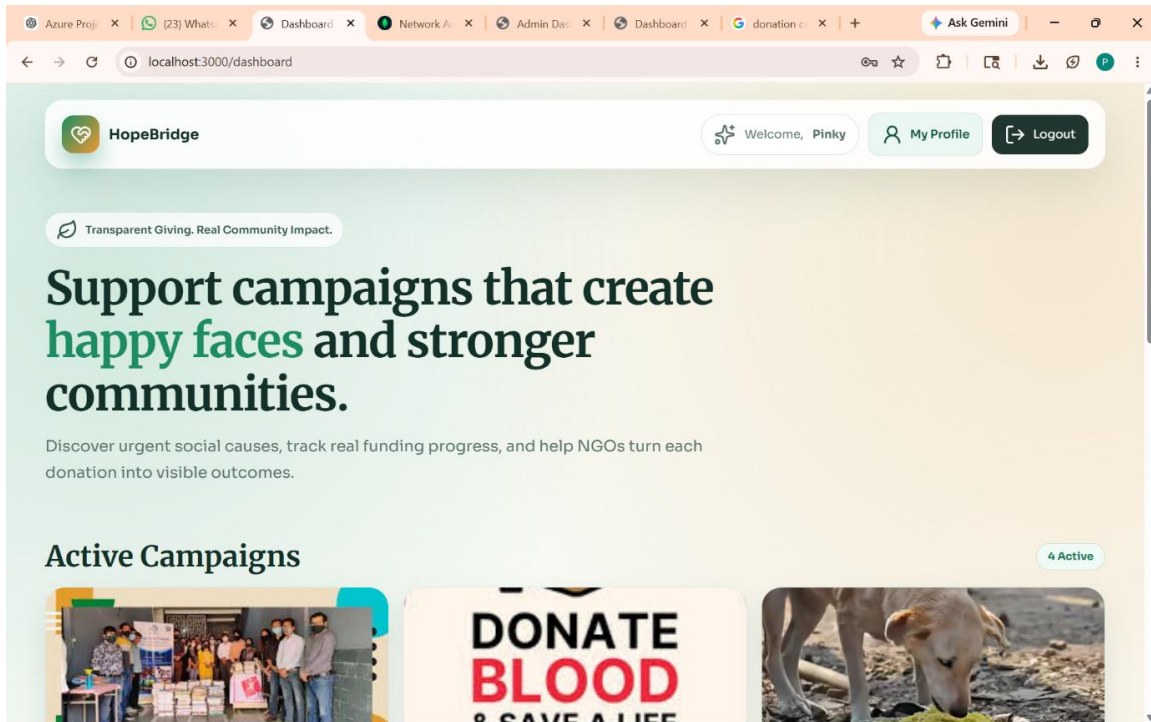
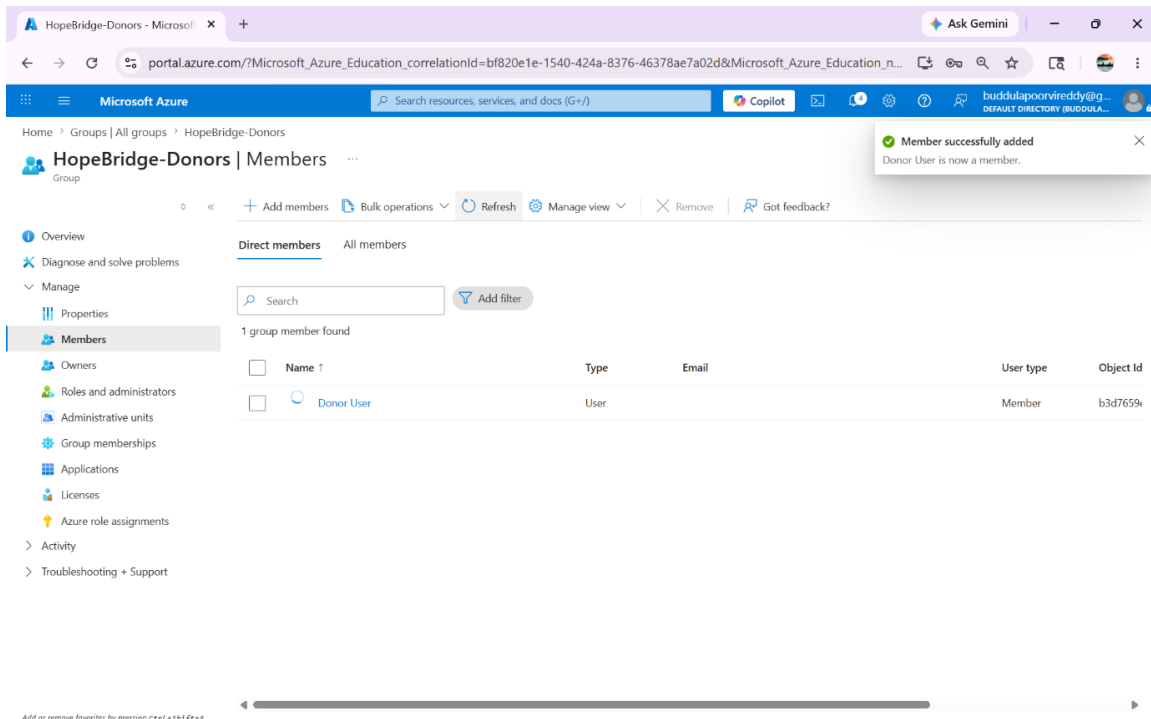
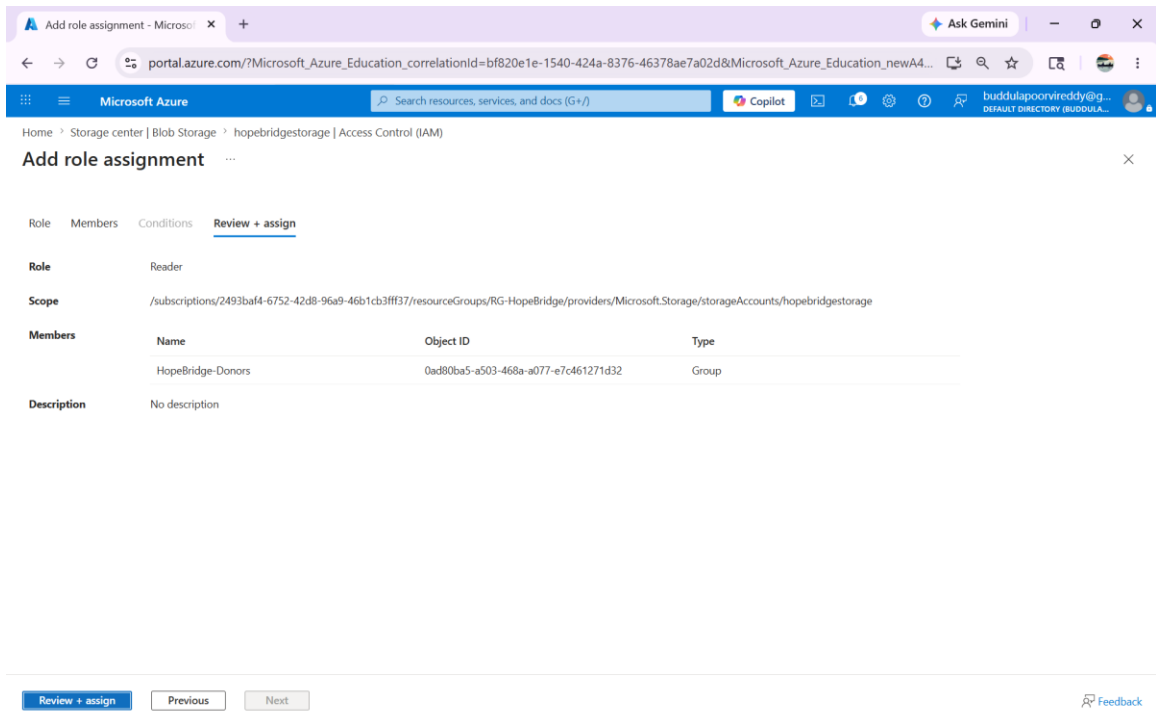
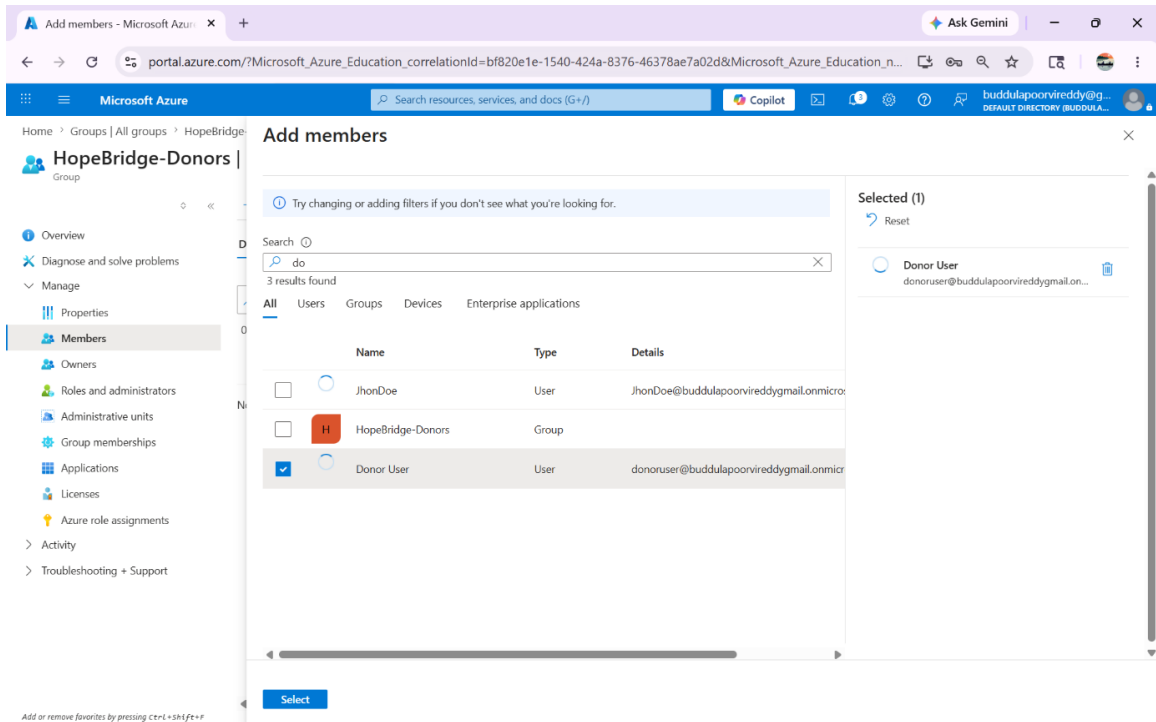


Figure 3: Donar Dash board





This figure demonstrates the implementation step related to blob container creation in the HopeBridge cloud project.

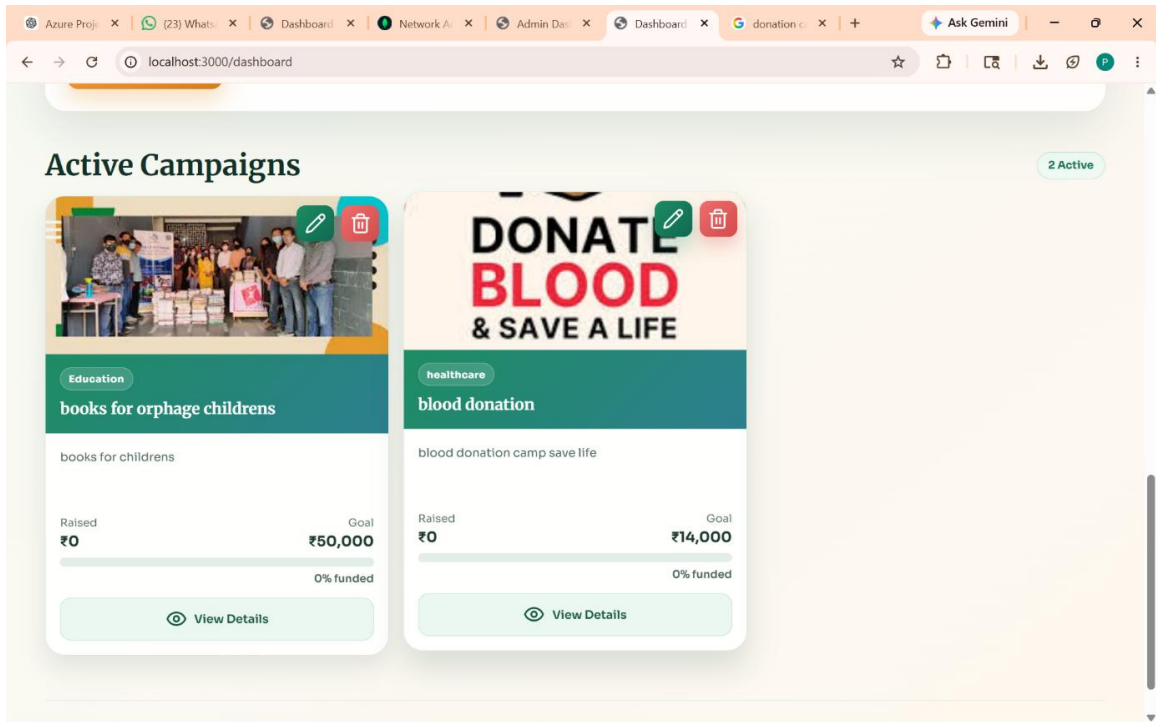


Figure 4: Blob Storage Configuration

Create a storage account - Mic

Ask Gemini

portal.azure.com/#view/Microsoft.Azure.Storage/CreateStorageAccount.ReactView/_provisioningContext~/7B"initialValues%3A%7B"subs...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

buddulapoorvireddy@ig...
DEFAULT DIRECTORY (BUDDULA...

Home > Storage center | Blob Storage

Create a storage account

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below.
[Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * Azure for Students

Resource group * RG-HopeBridge
[Create new](#)

Instance details

Storage account name * hopebridgestorage

Region * (Asia Pacific) Central India
[Deploy to an Azure Extended Zone](#)

Preferred storage type Azure Blob Storage or Azure Data Lake Storage Gen 2

Performance * Standard: Recommended for most scenarios (general-purpose v2 account)
Premium: Recommended for scenarios that require low latency

Previous Next Review + create

[Give feedback](#)

hopebridgestorage_177870350

Ask Gemini

portal.azure.com/#view/Microsoft.Azure.Resources/DeploymentDetails.MenuView/~/_overview/id/%2Fsubscriptions%2F2493baf4-6752-42d...

Microsoft Azure

Search resources, services, and docs (G+/)

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DEFAULT DIRECTORY (BUDDULA...

Home

Deployment

hopebridgestorage_1778703500008 | Overview

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : hopebridgestorage_1778703500008 Start time : 14/05/2026, 01:49:38
Subscription : Azure for Students Correlation ID : 13db1f70-c76b-4527-9eb3-d440d0db249e
Resource group : RG-HopeBridge

Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Microsoft Defender for Cloud

Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

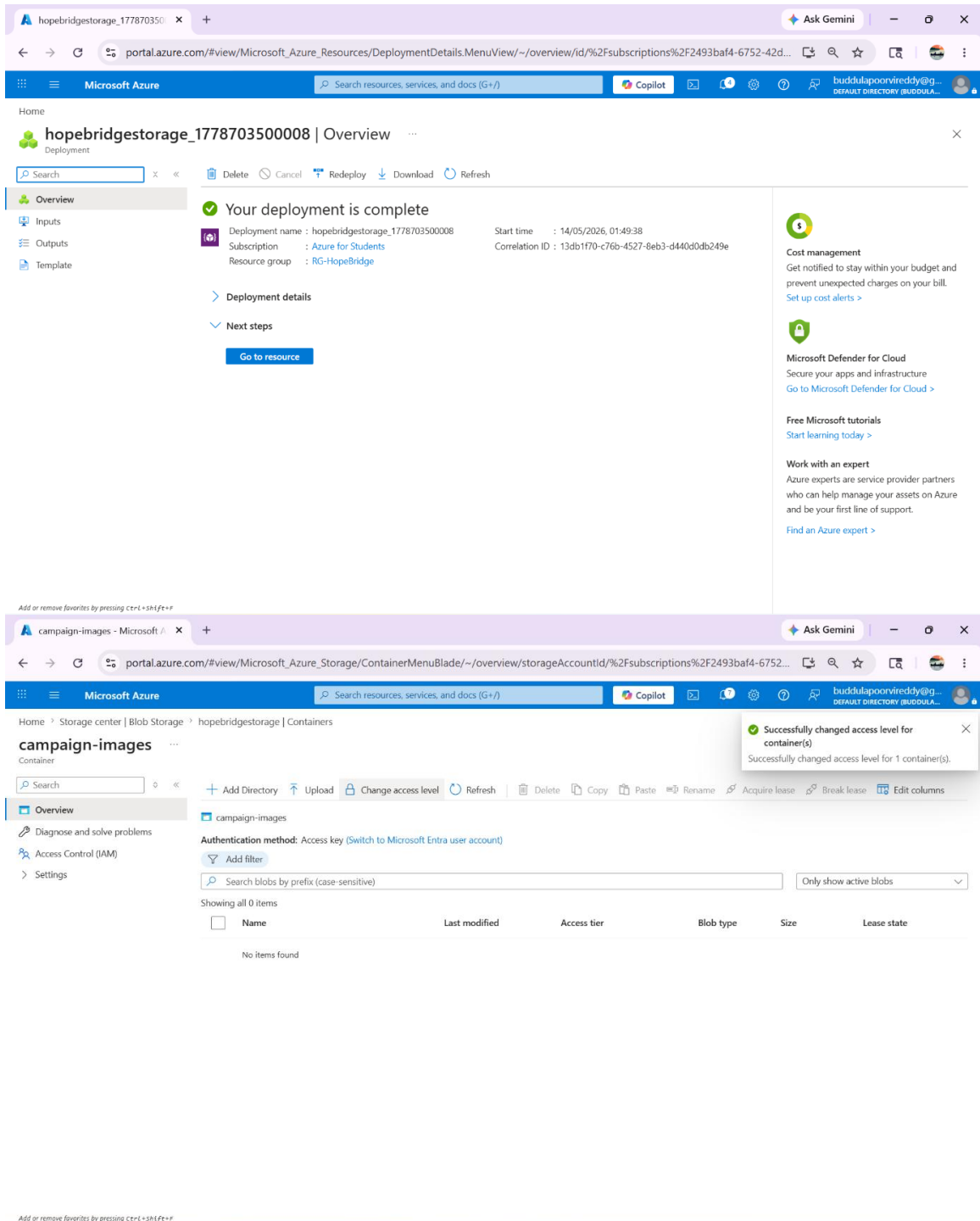
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This figure demonstrates the implementation step related to blob storage configuration in the HopeBridge cloud project.

Create A New Campaign

CAMPAIGN TITLE
blood donation

GOAL AMOUNT (₹)
14000

DEADLINE
26-05-2026

CATEGORY
healthcare

CAMPAIGN IMAGE (OPTIONAL)
Choose file images (1).jpg

CAMPAIGN DESCRIPTION
blood donation camp save life

Launch Campaign

Figure 5: Campaign Image Upload in Azure Blob Storage

Microsoft Azure

campaign-images

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	1778773280149-88879011.avif	14/05/2026, 21:11:21	Hot (Inferred)	Block blob	11.67 KiB	Available

This figure demonstrates the implementation step related to campaign image upload in azure blob storage in the HopeBridge cloud project.

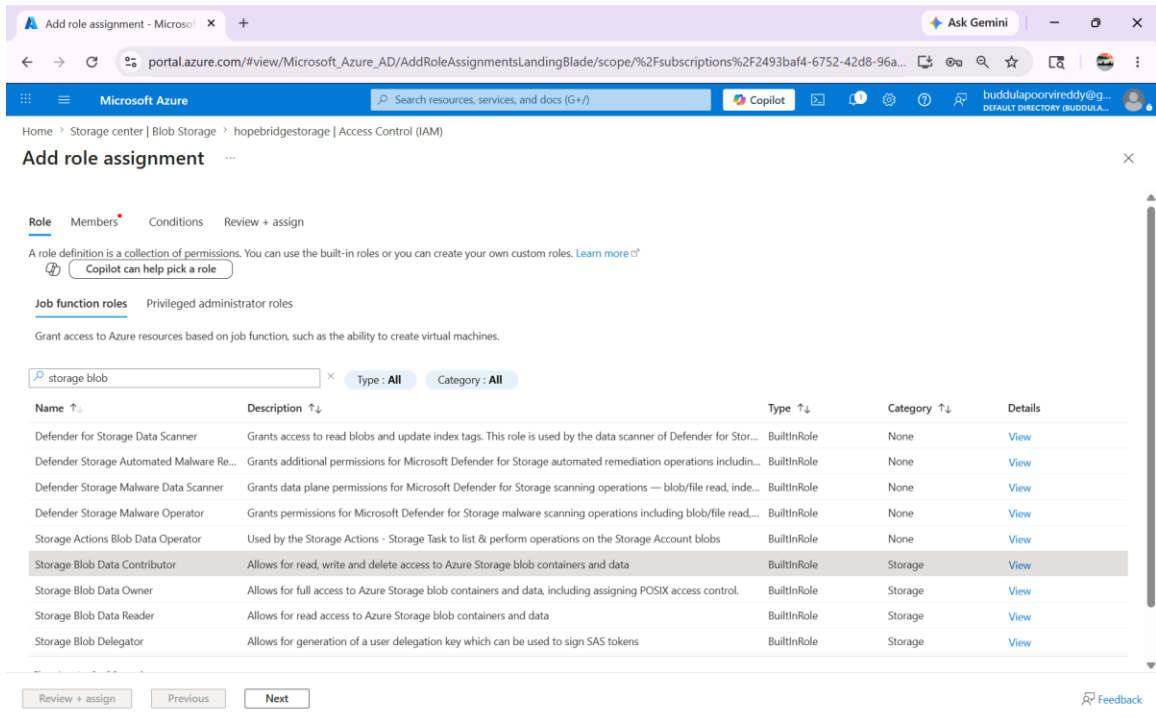


Figure 6: here is to assign the roles for users

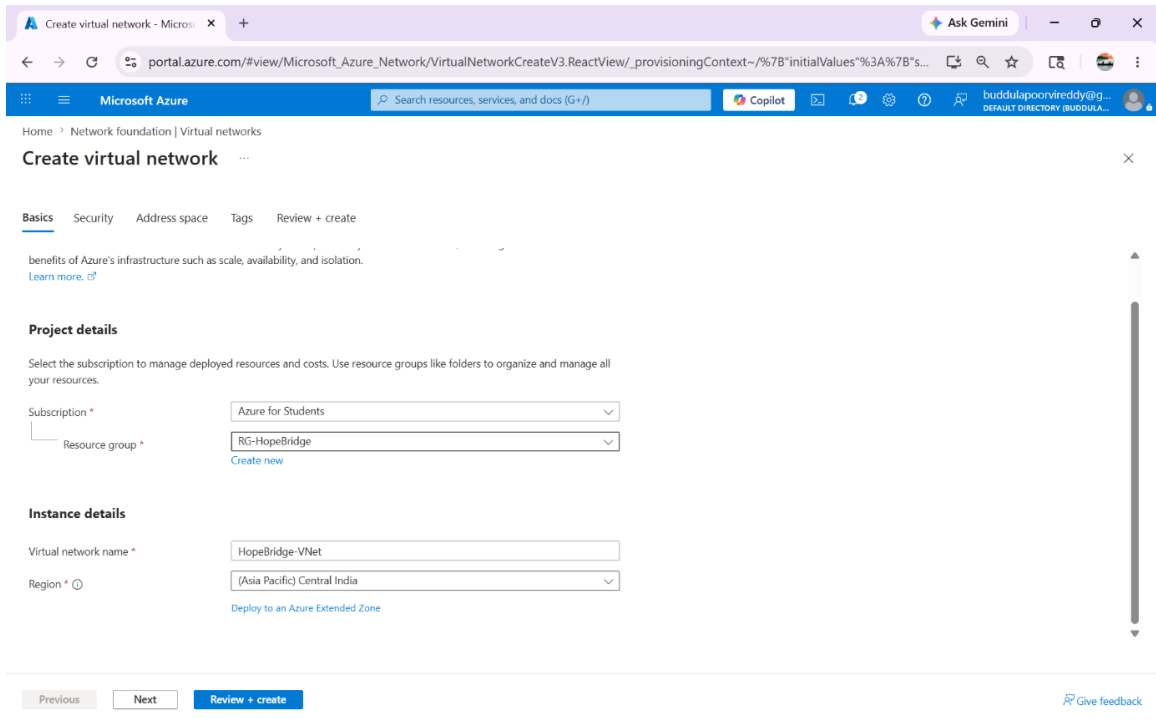


Figure 7: here is Vnet creted and NSG is creted along with subnet

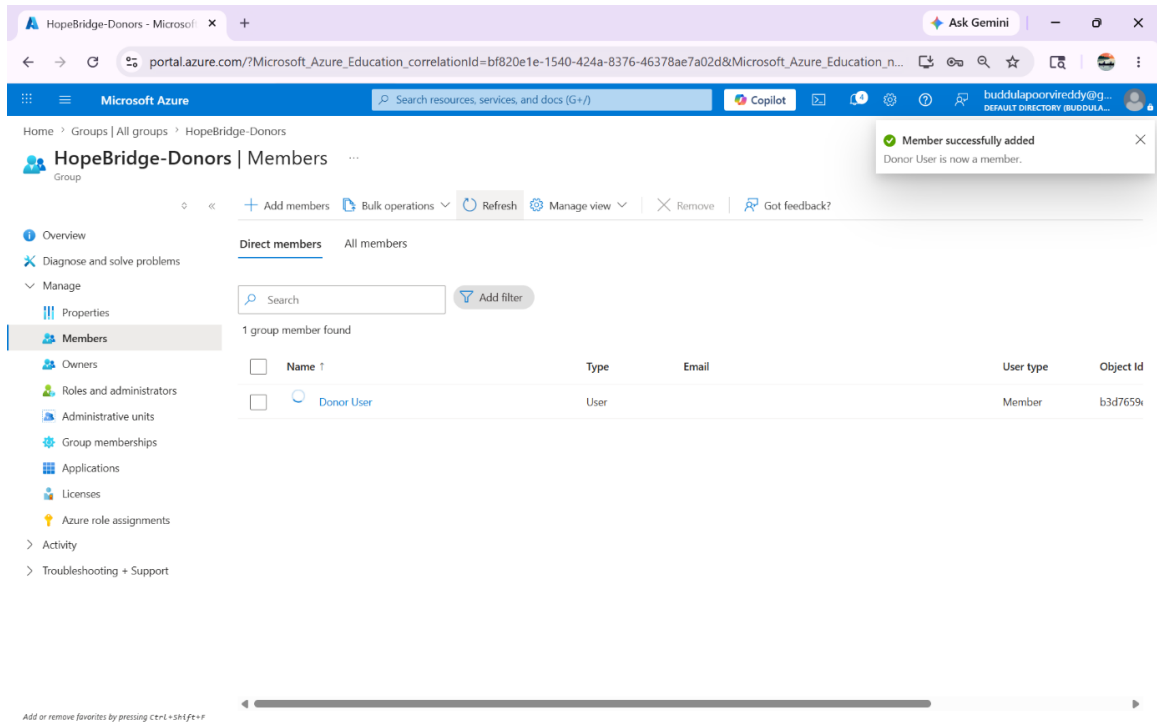


Figure 8: Azure Entra ID User Creation

This figure demonstrates the implementation step related to azure entra id user creation in the HopeBridge cloud project.

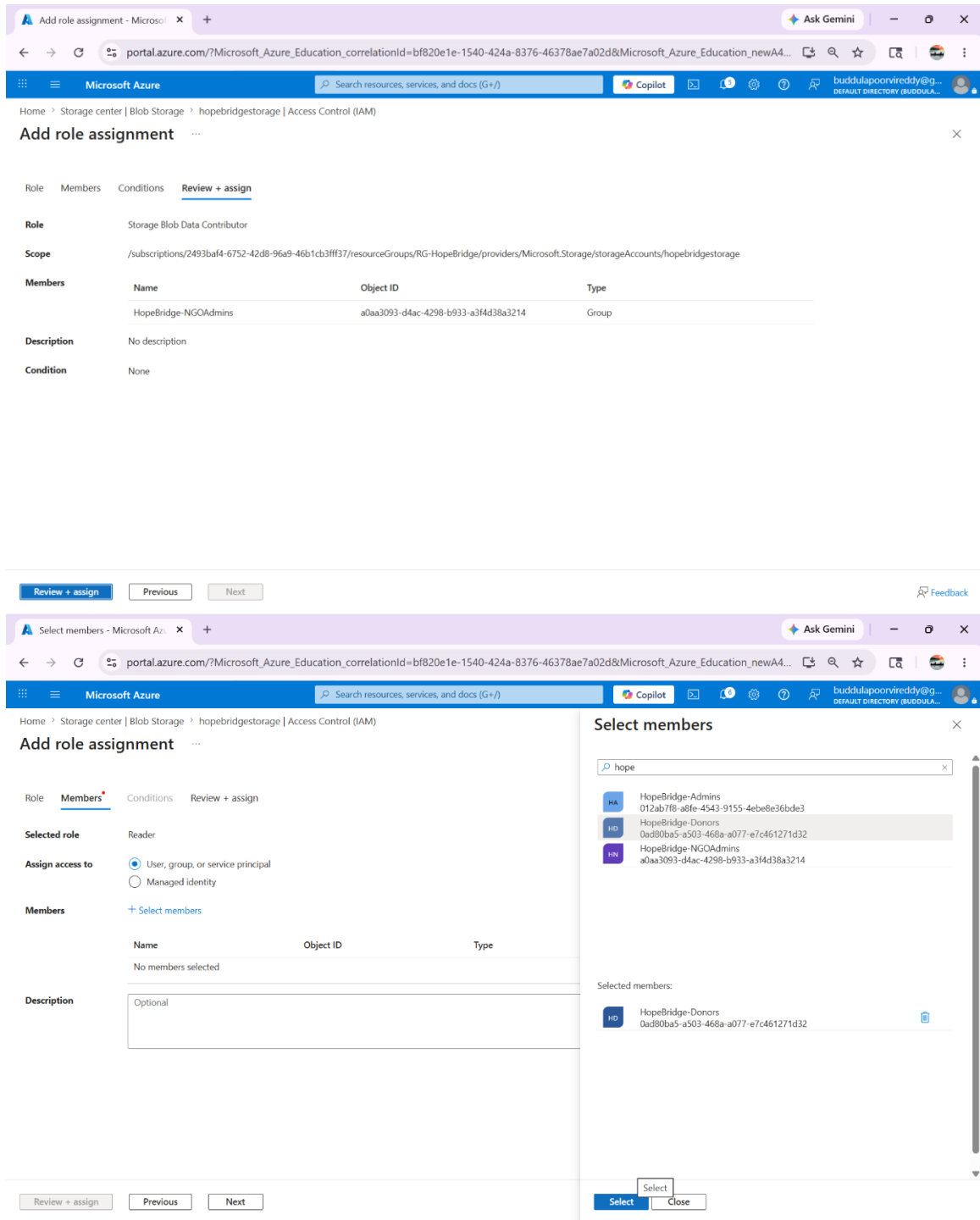


Figure 9: Azure Role Assignment using RBAC

This figure demonstrates the implementation step related to azure role assignment using rbac in the HopeBridge cloud project.

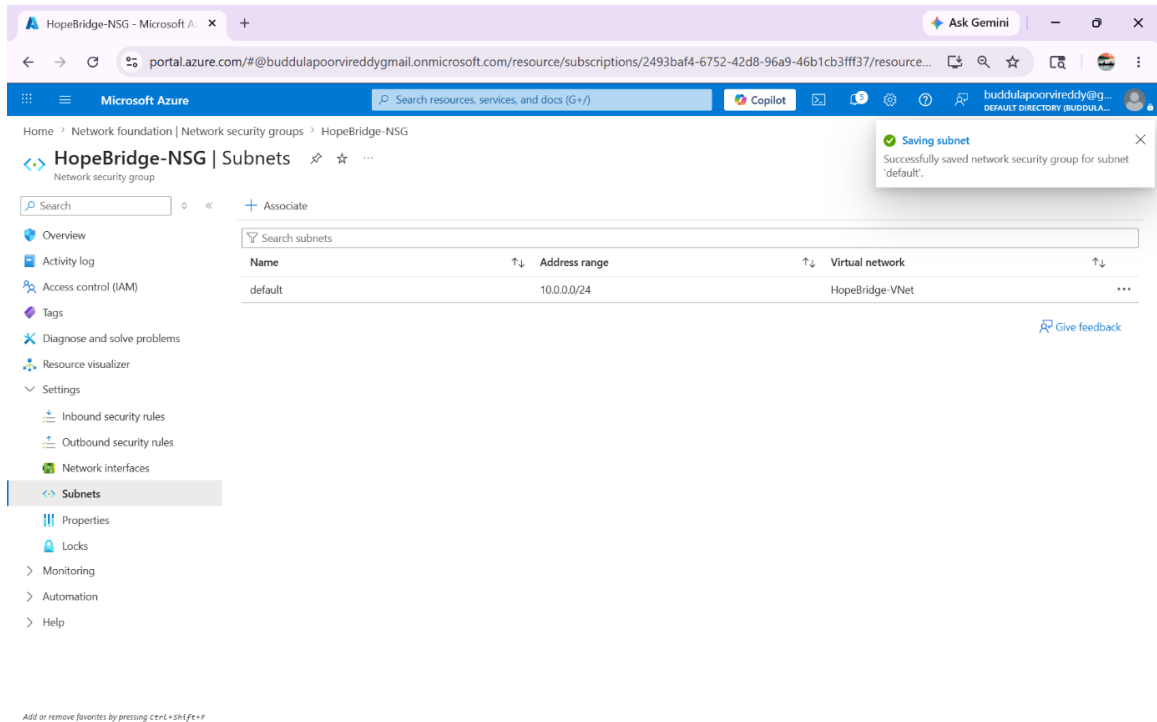


Figure 11: Network Security Group Configuration

This figure demonstrates the implementation step related to network security group configuration in the HopeBridge cloud project.

hopebridgestorage - Microsoft | portal.azure.com/?Microsoft_Azure_Education_correlationId=bf820e1e-1540-424a-8376-46378ae7a02d&Microsoft_Azure_Education_newA4...

Microsoft Azure | Search resources, services, and docs (G+)

Home > hopebridgestorage

hopebridgestorage | Diagnostic settings

Storage account

Subscription: Azure for Students | Resource group: RG-HopeBridge | Resource type: Storage accounts | Resource: hopebridgestorage

Diagnostic settings are used to configure streaming export of platform logs and metrics for a resource to the destination of your choice. You may create up to five different diagnostic settings to send different logs and metrics to independent destinations. [Learn more about diagnostic settings](#)

Name	Storage account	Event hub	Log Analytics workspace	Partner solution	Edit setting
No diagnostic settings defined					
+ Add diagnostic setting					

Click 'Add Diagnostic setting' above to configure the collection of the following data:

- Transaction

hopebridgestorage - Microsoft | portal.azure.com/?Microsoft_Azure_Education_correlationId=bf820e1e-1540-424a-8376-46378ae7a02d&Microsoft_Azure_Education_newA4...

Microsoft Azure | Search resources, services, and docs (G+)

Home > hopebridgestorage

hopebridgestorage | Metrics

Storage account

Local Time: Last 24 hours (Automatic - 5 minutes)

Sum Ingress for hopebridgestorage

+ Add metric | Add filter | Apply splitting | Line chart | Drill into Logs | New alert rule | Save to dashboard

Scope: hopebridgestorage | Metric Namespace: Account | Metric: Ingress | Aggregation: Sum

CAPACITY: Used capacity
TRANSACTION: Availability, Blob Geo-Replication Lag, Egress, Ingress, Source: F3E-1 status

Ingress (Sum), hopebridgestorage | 44.5KB

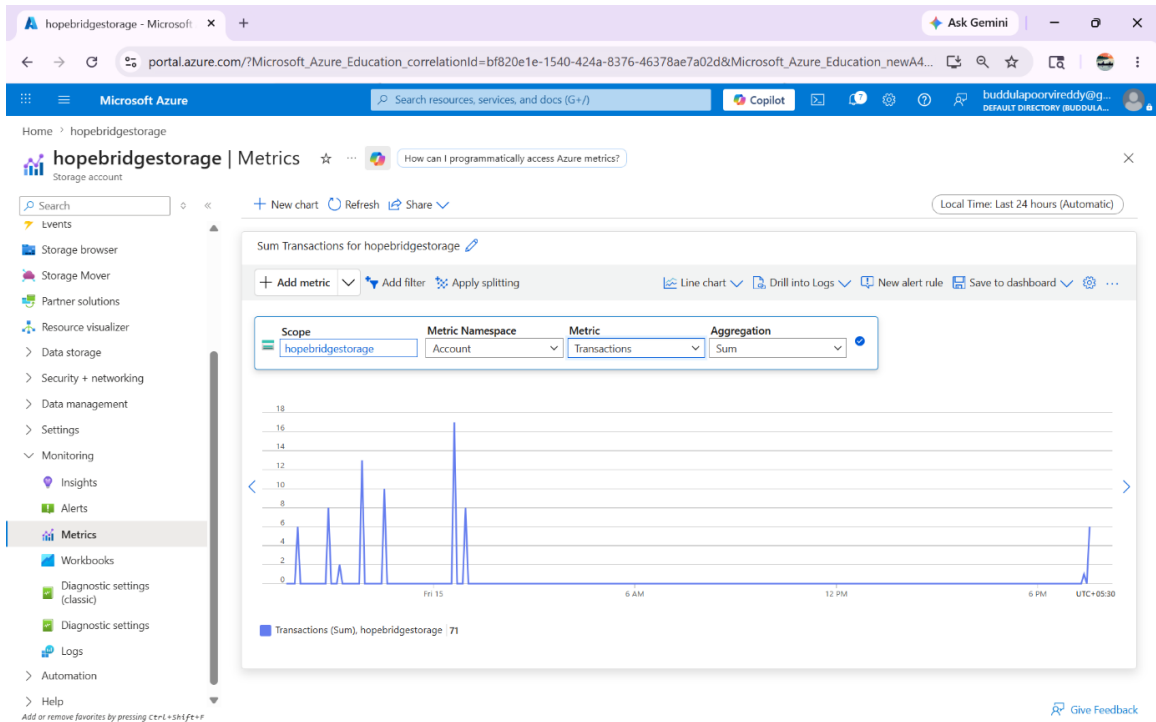


Figure 12: Azure Monitoring and Metrics

This figure demonstrates the implementation step related to azure monitoring and metrics in the HopeBridge cloud project.

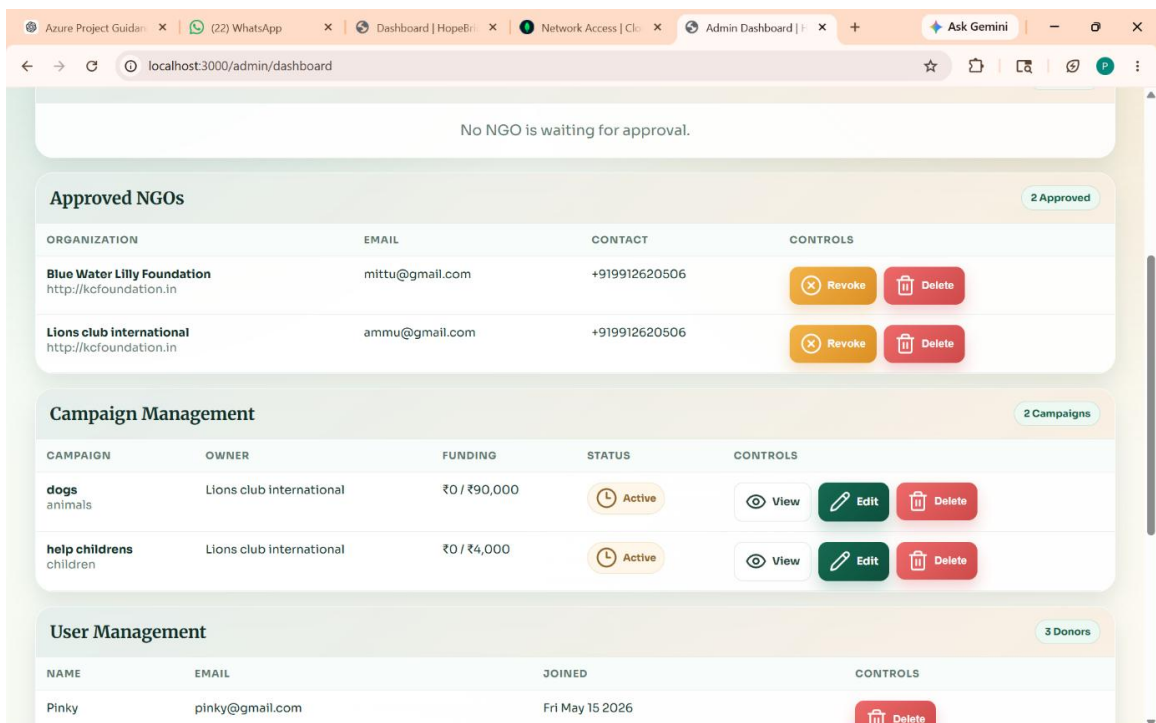


Figure 13: Admin Dashboard

This figure demonstrates the implementation step related to admin dashboard in the HopeBridge cloud project.

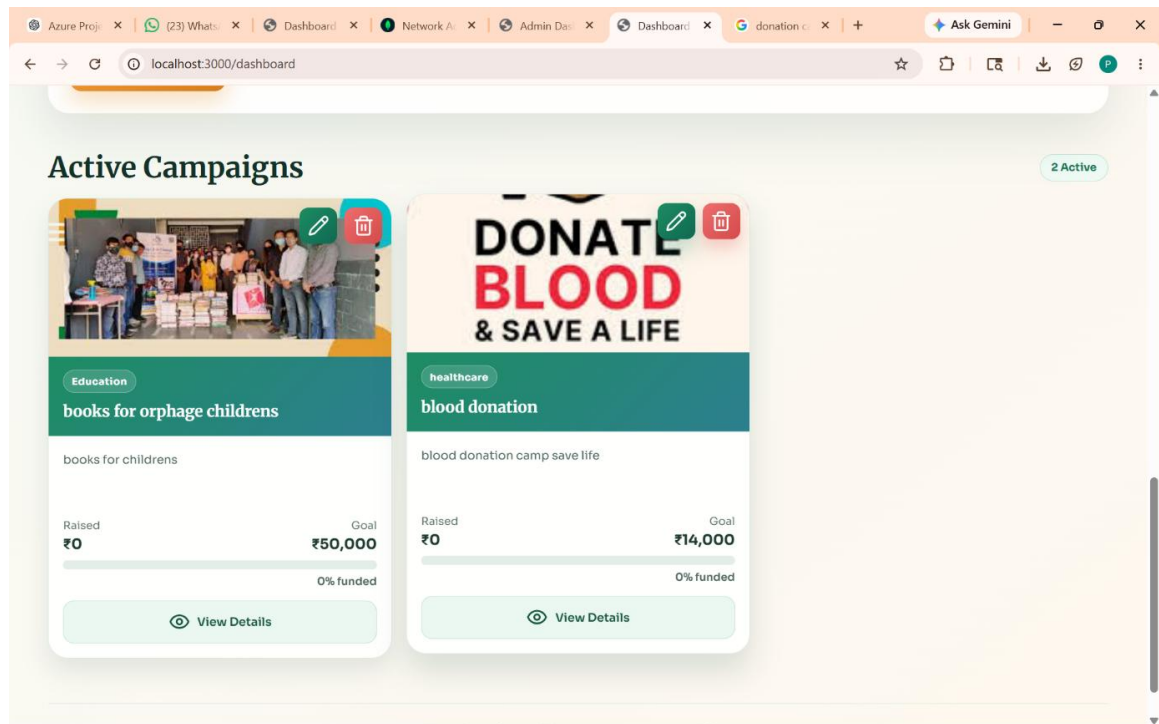


Figure 14: NGO Dashboard

This figure demonstrates the implementation step related to ngo dashboard in the HopeBridge cloud project.

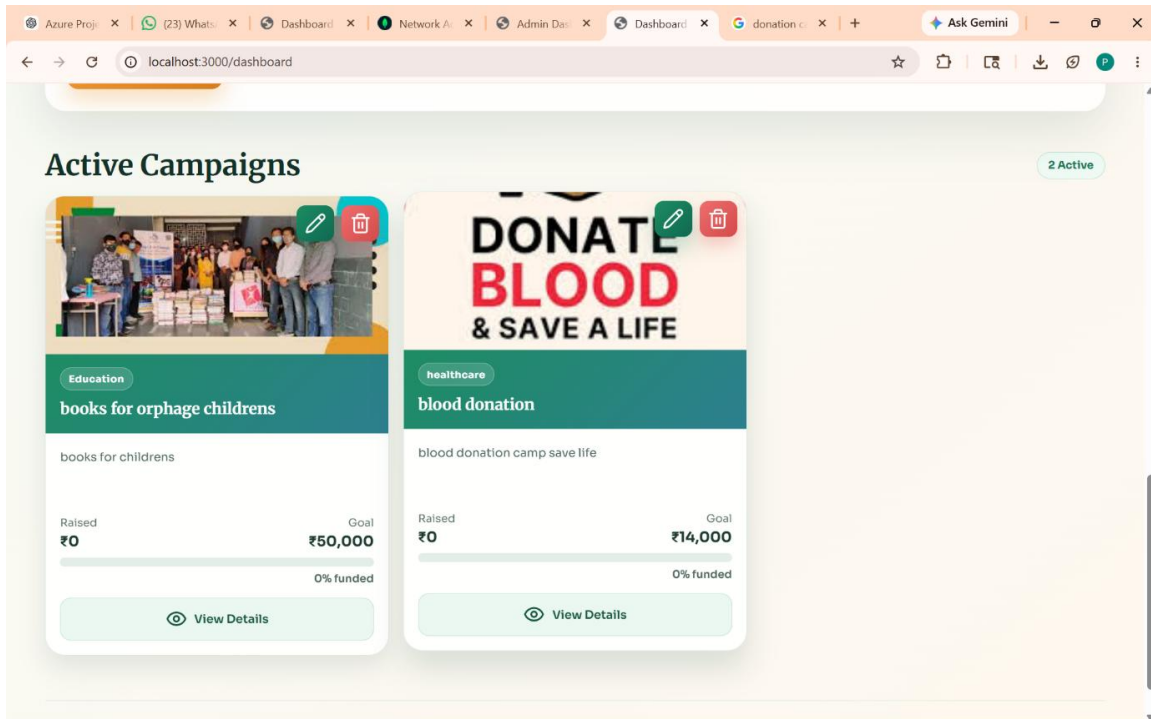


Figure 15: Donor Dashboard

This figure demonstrates the implementation step related to donor dashboard in the HopeBridge cloud project.

6. CONCLUSION

The HopeBridge platform successfully demonstrates the use of Microsoft Azure cloud technologies for building a secure and scalable NGO donation platform. The project integrates cloud storage, identity management, RBAC, networking, and monitoring services to provide a complete cloud-based solution.

7. REFERENCES

Microsoft Azure Documentation – <https://learn.microsoft.com/azure/>

Render Deployment Documentation – <https://render.com/docs>

Node.js Documentation – <https://nodejs.org/>

MongoDB Documentation – <https://www.mongodb.com/docs/>